Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/17/2022
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	SCALE: Shared Cognitive Architectures for Long-term Exploration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dechurch@northwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	954-646-5083
Organization Name:	Northwestern University		
PI Address 1:	Northwestern University		
PI Address 2:	School of Communication		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60208	**Congressional District:**	9
Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	11/09/2017	**End Date:**	06/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 10/21/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University) Johnson, Jeffrey Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC18K0221		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "SCALE: Shared Cognitive Architectures for Long-term Exploration," grant NNX15AM26G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. Among the remarkable team challenges NASA faces in long distance space exploration (LDSE) missions is the need to maintain team shared mental models (SMMs). Maintaining team SMMs requires the ability to detect shifts in cognition that will likely occur during the mission that could lead to ineffective crew functioning and performance. Maintaining team SMMs also requires validated countermeasures for bringing team members' cognitive understanding back into alignment. Leaving low Earth orbit is extreme teamwork -- team SMMs need to be maintained within teams operating close up (the crew), and between teams operating at an unprecedented distance (i.e., the crew & ground; 33 million miles in the case of a Mars Mission). A multidisciplinary research team will leverage expertise in Psychology, Industrial Engineering, & Anthropology to understand the emergence and outcomes of critical shifts in team cognition over LDSE missions. What are the triggering events of SMM divergence, how can we detect them, and which countermeasures most effectively bring them back into alignment? This project leverages a novel conceptual framework of shared cognitive architecture (SCA) to understand the patterns of SMMs that dynamically link members of teams, and teams to other teams, as they go beyond low Earth orbit. We use semantic analysis to identify cognitive shifts, and relational event network analysis to understand the antecedents and consequences of these shifts. We use these alongside an agent-based model fit on LDSE analogue data, so that we can explore an exhaustive set of potential triggering conditions that must be unpacked to conduct efficient ground analogue research. We then conduct this research in HERA (Human Exploration Research Analog), Moonwalk, and Antarctica. The project culminates in the evaluation of a dashboard fed with the results of computational modeling, human validation, and lexical markers to detect and suggest countermeasures to maintain SMMs through time and space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the environmental conditions faced by crews during LDSE missions are not unique to space travel. With our research program, we hope to reveal how conditions seen in organizations across the world impact shared cognitive architecture, and what we can do to mitigate these risks. Our meta-analysis has begun to investigate the impact of shared cognitive architecture on team outcomes across a variety of environmental conditions seen in LDSE missions that can also be applied to work in all organizations.
Task Progress:	Executive Summary of Phase 1: Through our Phase 1 efforts, we make three contributions to team risk in long distance space exploration missions. First, we developed a computational model that successfully predicts crew shared mental models (SMMs) based on personal, task, social, and situational factors. This model can be used in mission planning and mission support. Second, our comprehensive meta-analysis of prior research establishes that shared mental models are a robust predictor of team performance. Though we found moderators, the relationship between shared mental models and performance is always positive. The implication being that long distance missions should monitor and promote shared mental models among the crew, among mission support, and between the crew and mission support. Third, our work on the Skylab mission demonstrates that crew shared mental models, and shared mental models between the crew and ground can be assessed using text analysis of unstructured text, in this case we used mission transcripts of spoken communication. Executive Summary of Phase 2: Through our Phase 2 efforts, we make three contributions to team risk in long distance space exploration missions. First, we finalized the development of the computational model operationalizing shared mental models as network ties among crewmembers and we ran “what-if” computational experiments to explicate how leadership network structures affect mental model development in teams over time. Fitting the empirical data to the computational model showed that crew relations (e.g., leadership), crew composition (e.g., cognitive ability, agreeableness diversity), and mission task design (e.g., workload, task importance) are consistent predictors that could be targeted for SMM countermeasure development. Our second contribution is to use the model to identify particular configurations of relations that undermine shared mental models. Results of computational experiments suggest that shared leadership results in the most shared mental models, followed by hierarchical leadership, and then connected leadership; factionalized and disenfranchised leadership structures result in the least shared mental models. This finding suggests informal leadership is a useful crew property to monitor, and that the formation of factions, or occurrence of disenfranchised crewmembers should trigger crew support countermeasures as these structures in the crew pose a threat to mental models and therefore, performance. The third contribution is to explain how crew and ground schemas about communication networks affect information sharing. We found crewmembers tend to overestimate the presence of communication links between members who are part of the same formal group, and underestimate the presence of links between people who are in different groups. This deliverable underscores the importance of accurate relational schema, one aspect of mental models, as a precursor to effective information sharing and performance. Executive Summary of Phase 3: Through our Phase 3 efforts, we make three contributions to team risk in long distance space exploration missions. First, we developed a prototype dashboard for the detection of shared cognitive architecture. Second, we explored the correlation between shared mental model and team performance in the NASA Human Exploration Research Analog (HERA). We found substantial positive correlations of shared mental models and both dimensions of team performance in HERA. Though shared mental models are a strong predictor of team performance across mission stages, we found some nuanced shifts: (1) shared mental models are the most strongly predictive of performance on creative thinking tasks before and after communication delay; (2) on problem solving tasks, shared mental models increase in predictive power continuously throughout the mission; (3) of all the outcomes, fluency is the most predicted by shared mental models. Second, we examined how isolation, confinement, and network position affect network routing accuracy. Using a network routing task, we found that network acuity among HERA crews was significantly higher than mission control members in C4 and C5, but not C3. Additionally, we found communication delays did not impact network acuity among HERA crews, but reduced acuity among mission control members. Further, crewmembers who scored lower on the personality characteristic of conscientiousness had higher network acuity in both C3 and C4. Finally, crews’ network acuity was associated with personality characteristics of openness to experience, agreeableness, and neuroticism in C4, but not C3. Overall, our findings suggest that selecting crews with high network acuity will play a key role in alleviating the risk of information-sharing failures within a multiteam system under conditions of communication delays. Note: See the FY 2021 report for additional information, including a year-by-year summary of this project (Ed., 10/14/22).

Bibliography Type:	Description: (Last Updated: 11/17/2025)
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